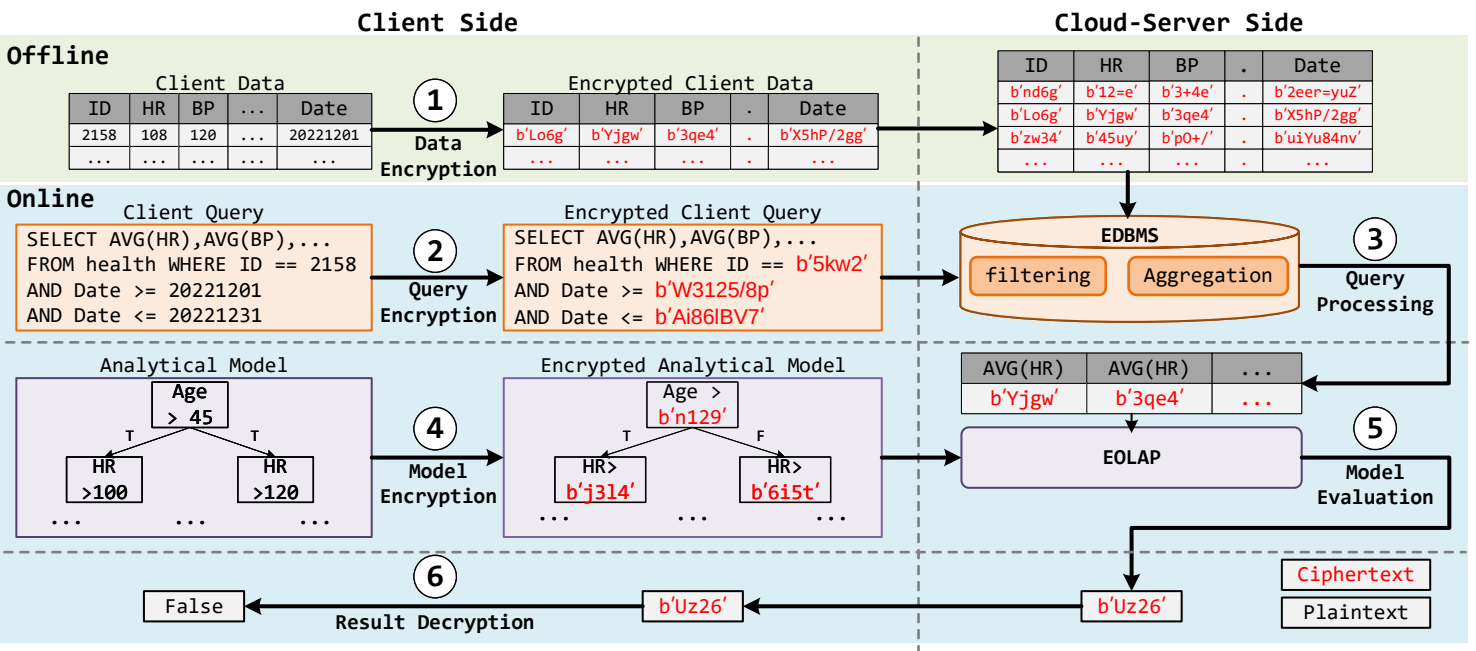

Client Side
Cloud-Server Side
Offline
Client Data
ID HR BP ... Date
2158 108 120 ... 20221201
... ...
1
Data Encryption
Encrypted Client Data
ID HR BP . Date
b'Lo6g' b'Yjgw' b'3qe4' . b'X5hP/2gg'
... ...
ID HR BP . Date
b'nd6g' b'12=e' b'3+4e' . b'2eer=yuZ'
b'Lo6g' b'Yjgw' b'3qe4' . b'X5hP/2gg'
b'zw34' b'45uy' b'pO+/' . b'uiYu84nv'
... ...
Online
Client Query
SELECT AVG(HR),AVG(BP),...
FROM health WHERE ID == 2158
AND Date >= 20221201
AND Date <= 20221231
2
Query Encryption
Encrypted Client Query
SELECT AVG(HR),AVG(BP),...
FROM health WHERE ID == b'5kw2'
AND Date >= b'W3125/8p'
AND Date <= b'Ai86lBV7'
EDBMS
filtering
Aggregation
3
Query Processing
Analytical Model
Age > 45
T
T
HR >100
HR >120
4
Model Encryption
Encrypted Analytical Model
Age > b'n129'
T
F
HR> b'j3l4'
HR> b'6i5t'
AVG(HR) AVG(HR) ...
b'Yjgw' b'3qe4' ...
EOLAP
5
Model Evaluation
6
Result Decryption
False
b'Uz26'
b'Uz26'
Ciphertext
Plaintext